The Influence of Virtual Character Actions and Emotional Expressions on Trust in Autonomous Vehicles

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Abstract

With the rapid advancement of Autonomous Vehicle (AV) technology, improving user trust is critical to realizing its full potential. This study introduces a Virtual Character (VC) as an assistant for passengers, designed to enhance trust in AVs. The VC acts as an interactive guide, providing feedback and emotional cues to capture passengers' attention during uncertain situations. To assess its impact on trust, Situation Awareness (SA), and acceptability, we integrated the VC into a Virtual Reality-simulated AV environment. Thirty-six participants experienced four conditions: a static VC, an emotion-only VC, an action-only VC, and a combined action–emotion VC.

Results show that incorporating a VC can enhance user trust, SA, and acceptance of AV technologies. VC actions have a stronger effect on trust than emotional expressions, while emotional cues alone have the potential to reduce stress. Notably, participants preferred a VC that combines both action and emotion. These findings highlight the value of integrating expressive and behavior-rich VCs

into AV interfaces to communicate system awareness, reduce uncertainty, and strengthen passenger trust.

Keywords: Augmented Reality, Virtual Reality, Virtual character, Human-machine Interaction

1 Introduction

Autonomous Vehicles (AVs) hold immense potential to transform transportation systems, promising more efficient and convenient mobility solutions [1, 2]. However, the widespread adoption of AVs hinges on establishing trust between users and these sophisticated technologies. Trust plays a pivotal role in influencing user acceptance, adoption, and ultimately, the success of AVs in the market [3]. Drawing from prior research, several factors have been identified as contributing to a lack of trust in automation, such as being unsafe [4], feelings of fear [5], decreased sense of control [6], and unpredictable machine behavior [7]. SAE International [8] defines five levels for self-driving cars, with Level 5 representing full driving automation. Such AVs lack a traditional driver, potentially inducing fear and diminishing passengers' sense of control during uncertain situations.

To bridge the communication gap between AV systems and passengers, researchers have explored the integration of avacars. The term avacars refers to avatars, virtual agents, and physical robotic embodiments that personify the AV and communicate with users through familiar humanlike social cues [9]. For example, Nissan[1] uses Virtual Reality (VR) to add a Virtual Character (VC) joining the trip with the passengers, and the avatar occupies a seat, to give them a sense it is are actually in the vehicle. Previous approaches, such as a virtual chauffeur that glances toward relevant objects [10] or non-humanoid robotic agents [11] have demonstrated benefits in increasing comfort and trust. However, these implementations are limited to specific actions or pre-trip interactions, constraining their ability to support passengers during dynamic, in-ride situations.

Our research is focused on increasing trust in an AV by adding a VC. The objective of this VC is to assist and guide passengers in understanding the AV's decisions and the surrounding situation, especially during uncertain situations. Uncertainty in the context of AVs pertains to unpredictable scenarios or conditions the vehicle might encounter while navigating its environment. These situations can be a pedestrian suddenly going across the street or other vehicles driving unpredictably. Passengers want to know the specific reasons behind the uncertainty [12]. As one of the key motivations for this study, our VC points to the source of uncertainty to convey, in a very easy-to-understand manner, that the AV will take appropriate action because it is aware of an incoming event and will respond accordingly. This intuitive communication mechanism is expected to enhance passenger confidence and engagement.

[1] Nissan: Invisible-to-Visible (I2V) - Innovation. Available at https://www.nissan-global.com/EN/INNOVATION/TECHNOLOGY/ARCHIVE/I2V/

We anticipate that to indicate uncertain situations, a VC could provide a more natural form of communication compared to arrows or audio cues. Additionally, VCs could potentially express emotions to calm the passengers. The VC's emotional expressions might convey emotional trust, and their actions could communicate the car's intentions and actions.

Augmented Reality (AR) presents a promising technology for implementing such VCs, as it allows virtual elements to be overlaid onto the physical world and spatially registers them to the physical environment [13]. Previous research has demonstrated that AR can improve trust in AVs and promote safety in AVs [14, 15].

To demonstrate and validate our new VC concept, we developed a system that emulates AR in a VR environment to create the AV experience. This VR-based simulation provides an immersive, realistic, and controlled setting, serving as a comprehensive platform for assessing the effectiveness of VC in fostering trust and enhancing the user experience. A user study is conducted in this simulated AV environment to explore the impact of the VC's actions and emotions on trust, Situation Awareness (SA), and the Technology Acceptance Model (TAM) [16]. Results from a user study indicate that using a VC with actions, emotions, or a combination of both significantly increases trust compared to a neutral VC. Furthermore, VC actions have a more positive impact on trust, SA, and likability when combined with emotional expressions

This work presents the following two contributions:

- The design and implementation of a VC that integrates specific actions, emotions, and appearances to enhance human trust in AVs. The VC communicates the AV's ability to recognize and manage dangerous situations by conveying its intentions to the user during uncertain scenarios.
- Investigated the impact of a VC's emotions and actions on human trust, SA, likability, and acceptability. Results from a user study indicate that using a VC with actions, emotions, or a combination of both significantly increases trust compared to a neutral VC. Furthermore, VC actions have a more positive impact on trust, SA, and likability when combined with emotional expressions.

In Section 2, the paper begins with a review of the concept of trust in automation, exploring factors that contribute to trust and examining how VC may play a role in fostering trust in automated systems. In Section 3, we then provide a detailed description of our proposed VC, outlining its features, intended interactions, and mechanisms for enhancing user trust. Next, in Section 4, we present our user study and its findings, focusing on participants' interactions with the VC and the resulting impact on trust levels. Section 5 discusses the experimental results, interpreting the data and its implications for trust-building and user engagement with automation. Finally, in Section 7, we conclude with remarks on our study's key insights, limitations, and directions for future research.

2 Related Work

2.1 Trust in AVs

According to Lee and See, trust is *"the attitude that an agent will help achieve an individual's goals in a situation characterized by uncertainty and vulnerability"* [17]. Trust in automation depends on the performance, purpose, and process (i.e., understanding how the system makes decisions) of the system [18]. Mayer and Davis [19] showed that the factors impacting the perceived trustworthiness are ability, predictability, and integrity, i.e., the trustor's perception that the trustee adheres to a set of principles that the trustor finds acceptable. Based on these two models of trust, Korber [20] proposed a questionnaire to measure Trust in automation with 14 items consisting of 6 sub-scales, measured by 5-point Likert scales.

Systems capable of explaining their reasoning are more likely to be trusted because they are easier for users to understand [21]. Petersen et al. [22] discovered that driver assistance systems enhancing SA can positively influence drivers' trust in AVs. When users trust the automated system, their stress and concerns diminish, enabling them to concentrate more effectively on secondary tasks. Taylor [23] proposed the Situation Awareness Rating Technique (SART) for assessing the Understanding of the situation provided, the amount of demand on attentional resources, and the Supply of attentional resources (perceived workload).

A framework for incorporating trust-related elements into the human-machine interface is presented by Ekman et al. [24]. This model addresses the nature and practice of system use in autonomous driving systems. This framework consists of three phases: the pre-use phase, the learning phase, and the performance phase. The pre-use phase is the events preceding the user's initial physical interaction with the AV system. The learning phase starts with the first interaction until the user is familiar with the AV system. The trust-affecting factors in the design of the human-machine interface (HMI) for this phase are feedback, the presence of uncertain information, why and how information, user customization, anthropomorphism (the system is made more human-like), and adaptation to user preferences [24]. The performance phase, which is viewed from a long-term usage perspective, usually occurs after the learning phase, although the two phases may sometimes occur simultaneously. During the test simulation, the learning phase is crucial, whereas the performance phase determines trust based on the system's performance and dependability.

2.2 Avacars to Support Trust in Human-Machine Interaction

The concept of the avatar has traditionally been defined as a "digital representation of a human user that facilitates interaction with other users, entities, or the environment" [25]. However, this view has been challenged. Ratan et al., [9] argue that such definitions are too narrow and exclude physical embodiments, such as robots, that can serve as communicative mediators. He proposed the term *avacar* to refer to both digital and physical representations that act on behalf of a human or system, facilitating interaction with users and the environment. These avacars may be displayed on screens, projected through AR interfaces, or embodied physically as dashboard robots.

By mimicking human behavior, avacars aim to restore transparency and emotional engagement in the human–AV interaction.

Several studies have examined the potential of avacars to enhance user trust and system transparency.

For instance, Chakravarthi et al., [11] introduced a non-humanoid physical robotic object positioned on the dashboard of the vehicle. This robot comprises an organic-shaped structure capable of horizontal rotation, with a circle positioned atop it to enable vertical movement. The researchers demonstrated that the robot's non-verbal behaviors could convey social cues, thereby enhancing trust and fostering a sense of safety compared to a static object in an AV. Participants perceived the robot as providing companionship. However, this was a non-humanoid robot, so it could not convey action or emotion like a humanoid robot or humanoid agent; this implementation was limited to pre-trip interactions.

Research demonstrates that anthropomorphic design elements influence user trust and acceptance of automotive technology [24, 26]. Other studies have explored more anthropomorphic designs to bridge the communication between AVs and users. For example, Zihsler et al., [27] developed Carvatar, a robotic agent mounted on the dashboard with an abstract human head shape. The device could rotate 360 degrees horizontally and 40 degrees vertically to mimic the head movements and gaze behaviors of a human driver. Carvatar was designed to direct its gaze toward traffic participants, signs, or other relevant objects, thereby indicating recognized objects and the car's awareness. These social cues are intended to make the interaction more natural and aim to increase trust in AV. Anthropomorphism does not guarantee improved trust. Chen et al., [28] found that highly anthropomorphized avacar, particularly when physically embodied, could inadvertently reduce perceived control and trust. Participants reported feeling as though the agent was "another person" competing for control, which undermined comfort and confidence in the system. Interestingly, the study also found that virtual characters with high levels of anthropomorphism were perceived as more trustworthy compared to a physical robot. This suggests that the form and medium of embodiment (physical versus virtual) play a critical role in how anthropomorphic agents influence user trust.

Instead of using a physical character, Hauslschmid et al., [10] compared visualizations in an AV using HUD, including an anthropomorphic virtual chauffeur, a world in miniature, and a car's indicators as the baseline. In a dangerous situation, the virtual chauffeur glances at the event and displays an exclamation mark. While the virtual chauffeur increased trust, no significant difference was found between the chauffeur and the baseline. This indicates that the avatar's behavior alone may not be enough to enhance trust meaningfully. Moreover, because the chauffeur was positioned with its back facing the participants, its glances or head turns may have been insufficiently salient to attract user attention, and participants were unable to see its facial expressions. This limitation likely reduced the potential benefits of anthropomorphic cues. Hauslschmid et al. also reported that the chauffeur primarily improved perceptions of comfort and well-being rather than perceptions of system competence.

Commercial AV systems are also incorporating avacars into their interfaces. NVIDIA[2] released in-vehicle VC that are easily customizable and visualized on a 2D screen. These VCs can help make recommendations, book reservations, access vehicle controls, and provide alerts. This visualization is crucial for building trust between the AV and its passengers, ensuring that occupants can be confident in the AV system's perception. Participants were more likely to interact with the VC in the AR environment than using video conference systems (on a 2D screen like Google Meet) [29].

The next section explores the emotional impact of a VC on trust in automation.

2.3 Exploring Emotional Intelligence in Human-Machine Interactions

Ekman et al. [30] identified six basic emotions: happiness, anger, disgust, surprise, sadness, and fear. Christian Peter et al. [31] divided the emotion into two dimensions: valence and arousal. Valence encompasses the range from negative to positive emotion, while arousal denotes the intensity or activation level of an emotion, varying from low to high. Valence is a crucial dimension in the study of emotions. Traditionally, researchers have operated under the assumption that emotions can be classified into two distinct categories: positive or negative [32].

The facial appearance and expressions of a humanoid virtual character can significantly impact the perceived trustworthiness and people's decision-making behavior [33]. The VC's reliability is not more important than the attractiveness of facial appearance [34]. Incorporating trusting nonverbal behaviors, such as smiling, into a computer agent, the perception of that character's credibility and trustworthiness could be increased [35]. Demeure et al. [36] showed that emotions displayed by agents can result in a higher perceived believability.

2.4 Research Gap

Despite increasing interest in avacars, existing research has largely focused on limited pre-trip interactions or simple nonverbal cues such as glances and head rotations, offering little insight into how more expressive virtual characters can support passengers during dynamic and uncertain driving scenarios.

Research on emotional expression in virtual characters further indicates that both facial appearance and affective cues shape perceived trustworthiness, yet their roles in the context of AV uncertainty have been insufficiently examined. Moreover, the relative impact of action-based versus emotion-based cues on trust, SA, and acceptance has not been systematically investigated. To address these gaps, this study introduces a 3D anthropomorphic VC embedded within a VR-simulated AV and systematically compares action-only, emotion-only, and combined action–emotion behaviors across uncertain driving situations.

A VC was selected as the communication medium because it enables the AV to convey both informational and emotional cues through humanlike social behaviors

[2]NVIDIA Research Group: Danny Shapiro: NVIDIA DRIVE Concierge Brings Customized AI to Every Seat. Available at https://blogs.nvidia.com/blog/drive-concierge-ai-experience/

such as pointing, gaze, and facial expressions. In addition, commercial systems such as NVIDIA[3]'s in-vehicle platform already employs 2D virtual characters displayed on screens, highlighting the growing relevance of digital characters in vehicle–passenger communication. Our work extends this direction by investigating how a fully animated 3D VC situated on the dashboard affects confidence. We use the term VC specifically to distinguish our design from physical embodiments explored in prior work.

3 Methods

3.1 Research Question and Hypothesis

The current study builds on this prior work by focusing on in-trip interactions and real-time environmental awareness. By conveying real-time visual cues for sudden vehicle responses, such as braking for a pedestrian, and by using emotion to calm or reassure, our VC offers an advanced form of interactive support.

This foundational investigation also explores how basic VC actions and emotional expressions influence passenger trust, user experience, and technology acceptance in AVs, providing a basis for future, more nuanced VC designs. By assessing both action and emotion, this study contributes to a better understanding of the role of VCs in AVs.

Research Question: *How do the actions and emotions of a Virtual Character (VC) influence trust, SA, and user acceptance in AVs?*

We hypothesize the specific impacts of the VC's actions and emotions on SA and acceptance. In complex driving environments, we expect that:

H1: Using AR techniques to add a VC with action, emotion, or both will increase trust compared to the baseline (no emotion and no action).

H2: Action-based and emotion-based VCs will increase users' Situational Awareness.

VCs performing actions, such as pointing to a potential hazard, can more effectively direct passenger attention to critical information in the environment, enabling passengers to understand that the AV has identified the potential hazard. Emotional expressions, such as a calm or reassuring smile, are expected to engage users, retain their focus on the situation, and assure the user that the situation is under control [37]. By investigating both action-oriented and emotion-expressive VCs, we aim to understand how these characteristics contribute to users' comprehension of complex driving scenarios in AVs.

H3: Emotion-only VCs will increase the acceptance of autonomous vehicles.

Emotion-only VCs focus on conveying emotional states such as trust, reassurance, or empathy. Research indicates that facial expressions and nonverbal cues reproduced by virtual agents can provide empathetic support during communication tasks [38]. This emotional engagement could help to mitigate user concerns or hesitations about AV technology by fostering a sense of calm and support during the AV experience, thus improving overall acceptance.

[3]NVIDIA Research Group: Danny Shapiro: NVIDIA DRIVE Concierge Brings Customized AI to Every Seat. Available at https://blogs.nvidia.com/blog/drive-concierge-ai-experience/

To address our research question, we followed these methodologies: (1) VC Design, (2) VR Environment Setup, (3) Experimental Condition Design, and (4) User Study Setup (detailed in Section 4).

3.2 Virtual Character Design

The overall design of the VC is friendly, inviting, and provides high-level information to the user, see Figure 2. The VC is designed as a user-friendly chauffeur VC, inspired by the work of Hauslschmid et al. [10]. In their design, the chauffeur VC is displayed on the HUD and positioned with its back to the participant. When the car approaches an object or encounters a potentially hazardous situation, the VC glances or looks at the object to draw attention to it. However, this positioning prevents participants from seeing the VC's face, and relying solely on glances or looks may make it difficult for participants to fully understand what is happening in the environment. Our VC is placed on the dashboard of the car, facing the participant (see Figure 2). The facial expression is designed based on the happy Emoji with open eyes [39]. The VC attracts attention and indicates uncertain conditions to the user through a set of animated actions, such as pointing, head rotation, and body orientation.

The modeling and animating of the VC were informally tested by five people (four PhD students and one project manager, who were not members of the research team) about its appearance. All of them agreed that the model was cute and that it did not scare them when the VC smiled. They found the actions of the VC easy to understand. The VC was created and animated via Blender 3.4.

The remainder of the section will describe the rational of the design choices and an in-depth discussion of the details of the VC design.

3.2.1 Design Rational

To design a VC that enhances trust through action and emotion, we first addressed the question: *What visual appearance is suitable for a VC in an AV?* Lawson-Guidigbe et al. [40] identified five basic models for an agent's virtual appearance in AVs: realistic human, mechanical human, realistic animal, mechanical animal, and abstract. Their research revealed that human and mechanical-human models garnered more trust in highly automated vehicles compared to other models. However, while human-like avatars may enhance trust, the "Uncanny Valley" [41] effect must also be carefully considered. This phenomenon shows a sudden decline in familiarity and trust as anthropomorphism increases beyond a certain threshold. The attributes of anthropomorphic appearance can be categorized into two main factors: human similarity, which refers to the degree to which the appearance resembles a human, and morphological completeness, which describes the extent to which body parts are fully represented [42, 43]. Research by Wang et al. [43] demonstrated that users prefer virtual robots with medium human similarity and high morphological completeness. Based on these findings, we designed our VC to incorporate both medium human similarity and high morphological completeness to optimize user trust while avoiding the Uncanny Valley effect.

Winterberger et al. [44] compared the effect of human drivers with AVs; their passengers had to ride with either an AV, a male, or a female driver, and the whole driving process was automated. They found that there are no significant differences between the passenger groups. As a result, we decided to use VC with a neutral gender.

3.2.2 Design Choices

There are three main design choices for our VC. In this section, we will describe the rationale for the choice of the color of the VC, of its actions (animation), and its emotion.

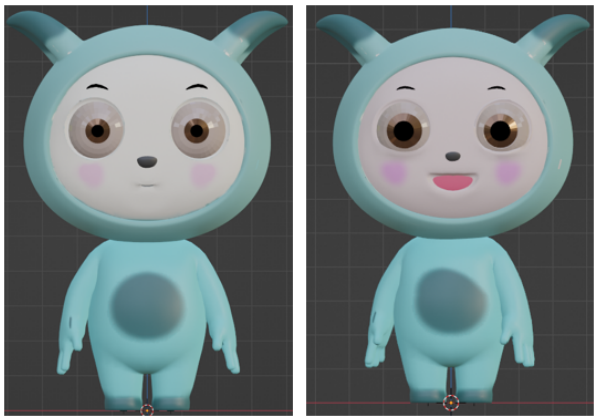
Color of the VC: Previous research showed that colors can impact the trust of consumers [45]. They demonstrate that blue is related more to trust than red. According to Mehta and Zhu [46], blue is commonly associated with peaceful and tranquil objects, such as the sky and the ocean, so it can reduce consumers' perceived risks of a trustee. Risk is an important part of trust; trust does not exist if a human does not perceive the risk of the AI model [26]. So if the blue color can reduce risk, it might also increase trust. Therefore, we chose to make the VC blue.

Action of the VC: In an AV, passengers may engage in secondary tasks such as reading. In dangerous situations, visualizations displayed via windshield displays or screens could pose a challenge in capturing passengers' attention. Prior research has attempted to signal environmental uncertainty through nonverbal behaviors such as head rotations or gaze shifts [10, 27]. However, head or eye movements alone can be difficult for passengers to notice. To address this limitation, our study integrates both head movement and explicit pointing gestures to create more salient cues. The VC is designed not merely as an information display but as a companion-like agent capable of drawing attention when necessary.

Colley et al. [47] used AR to incorporate information regarding the predicted trajectories of pedestrians and other vehicles via WSD. However, they discovered that this additional information did not lead to an increase in trust and instead resulted in a higher perceived workload compared to scenarios where no prediction information was provided. Displaying excessive information continuously may not yield optimal results. In our study, we aim to present information or objects only during uncertain situations selectively. The actions of our VC are as follows:

- Before the trip: the VC waves to welcome the passenger.
- During the trip: the VC stands to indicate attention to the environment (idle position).
- In dangerous situations: the VC attracts the passenger's attention through its actions, combining head and upper-body rotation with a clear pointing gesture toward the relevant object or event as soon as possible. This information will aid passengers in understanding the intentions of the AV and the situation around it, as well as confirm that the AV can safely manage the situation. We anticipate that these actions will increase SA and trust.

Emotion of the VC: Human interaction with a positive virtual agent, such as one displaying happiness, has a positive impact on the evaluation of the agent; our responses to a virtual agent closely resemble our reactions when interacting with



(a) Neutral VC (b) Smile VC

Fig. 1: (a) The neutral VC without emotional expression and action, (b) The VC with the smiling emotional expression.

humans [48]. According to Numata et al., [49], a positive response from a virtual agent is crucial in eliciting positive emotions in participants. Such a response can also help mitigate the effect of perceiving the agent as a computer program and facilitate effective human-virtual agent communication, particularly in terms of emotions [49]. Cowell and Stanney [50] suggest that the smile is arguably the most critical facial characteristic for conveying credibility. While we acknowledge the potential benefits of exploring a wider range of contextual emotions, our study initially focuses on evaluating the specific impact of smiling as an understood and positive emotional cue. We limited the number of conditions to make the experiment valid and under a time limit to reduce fatigue for the participants. Hence, we decided to incorporate smiling into the VC when uncertainty arises in the environment. By smiling, the VC may enhance passenger trust and convey the belief that the VC (representing the AV) can effectively manage the situation. As previously mentioned, we designed the VC with medium human similarity, to make the emotion of the VC simple, the emoji's emotion is used (see Figure 1). Research by Kaye et al. [51] indicated that emojis may be processed similarly to facial stimuli. Utilizing visual cues from facial expressions in emojis can minimize ambiguous interpretations [39].

3.3 Experimental conditions

Our research investigates the potential impacts of actions and emotions (see Section 3.2 for selection criteria) of a VC on trust, TAM, and SA in AVs. We designed four experimental conditions (see Figure 2):

- VC with only action: the VC uses head rotation and hand gestures to point to uncertainty in the environment (upper left Figure 2).
- VC with only emotion: the VC smiles when AV encounters an uncertain situation (lower right Figure 2).

- VC with action and emotion: the VC combines smiling with head rotation and hand gestures to point to uncertainty in the environment (upper right Figure 2).
- Static VC: Base condition with neither action nor emotion (lower left Figure 2).

To ensure a controlled comparison across conditions, we selected a baseline that included a static VC rather than an empty scene without a VC. This choice was motivated by Chakravarthi et al., [11]. A static VC keeps all visuals constant except the behavior being manipulated.

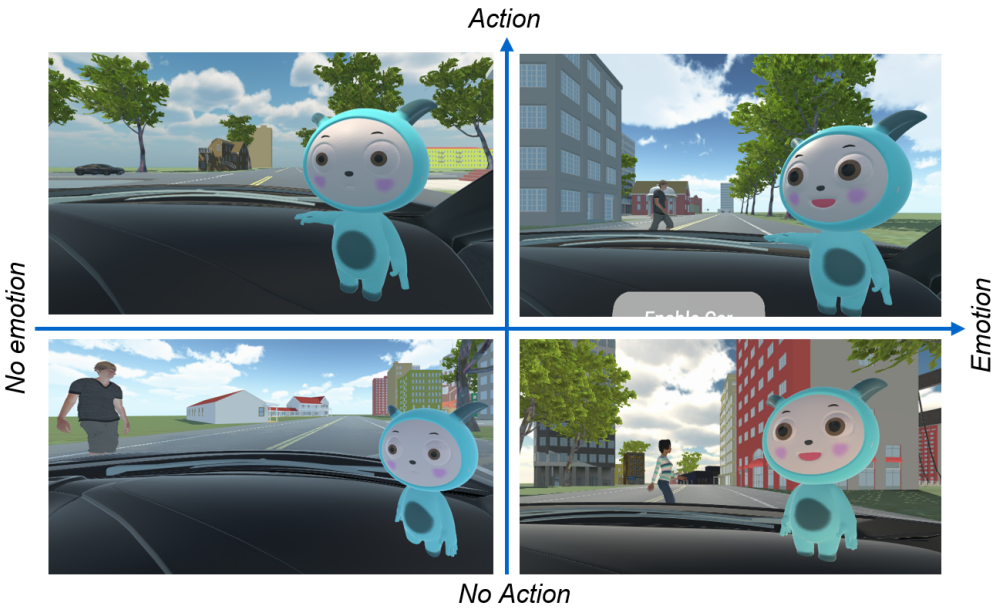


Fig. 2: Experimental conditions

3.4 Virtual Reality environment setup

AR has emerged as a promising technology in the automotive domain, offering new opportunities for enhancing driver and passenger awareness. Despite its promising potential, the integration of AR in vehicles presents several technical and perceptual challenges. These issues must be carefully addressed to ensure that AR systems enhance, rather than hinder, user experience and safety. According to Gabbard et al., [52], the challenges of in-vehicle AR include tracking and registration, depth perception and estimation, and a limited field of view.

VR driving simulators have the limitation of lower realism than physical driving due to the limited field of view or lack of motion feedback [53]. However, using a VR environment allowed us to maintain a high level of control over the experimental conditions. Due to the lack of existing fully automated vehicles on open roads in our country, we conducted our study in VR.

We used an AR simulation in VR technique to add a VC with only emotion (VCE), a VC with only action (VCA), a VC with action and emotion (VCAE), and a baseline (static VC). We utilized Unity3D version 2022.3.18f1 to build the VR-simulated AR environment, specifically designed to resemble a city setting.

We utilized the City package[4] from the Unity Asset Store, implementing four distinct paths for the four conditions to mitigate carry-over effects while maintaining an equal number of events across conditions. Streets were constructed using Road Architect[5], creating a comprehensive city environment for our autonomous driving simulator that includes tunnels, bridges, and varying road inclines and declines. For the experimental events, we designed scenarios with potential risks rather than actual accidents involving the AV. These included situations such as pedestrians suddenly crossing the street and encounters at unmarked intersections with approaching vehicles. The environment also incorporated standard traffic control elements: one traffic light and one stop sign.

The integration of AVs into VR shows significant potential and has already captured the interest of various companies in the automotive industry, such as Nissan and Huyndai [29]. The advantage of VR is that we can conduct simulations in a safe and controlled environment. This project was developed based on the open-source application for AV simulation in Unity3D: AutoWSD[6]. The Oculus Quest Pro and a computer equipped with an Nvidia GeForce GTX 3060 and 16GB of RAM were used.

3.5 Measures

To examine the impact of the VC's actions and emotions on trust in AVs, we analyzed the following qualitative data: Trust [20], TAM [54], SA [55], and Likability [16]. The four measures are as follows:

Trust: We use the trust in Automation questionnaire by Körber [20]. The questionnaire includes three subscales: Reliability/Competence, Understandability/Predictability, and trust. Risk is an essential component of a model of trust [19, 56], so we add the Perceived risk sub-scale to our questionnaire.

Situational awareness: We use SART [23] questionnaire with subscales Understanding and Supply to measure the SA of participants.

Acceptability: TAM is used to assess the acceptability of our VC visualization [16]. We use two subscales in this questionnaire to measure the participant's level of acceptability of the technology: perceived usefulness (PU) - *"the degree to which a person believes that using a particular system would enhance their job performance."* and perceived ease of use (PEOU) - *"the degree to which a person believes that using a particular system would be free from effort"*.

Likability: We also wanted to know if participants liked the VC or enjoyed the trip with the VC. Two questions, adapted from Waytz, Heafner, and Epley [57], were presented: "How comfortable did you feel?" and "How much participants would like to own a car like this one?".

[4] 255-pixel studios: CITY package. Available at https://assetstore.unity.com/packages/3d/environments/urban/city-package-107224

[5] MicroGSD: Road Architect for Unity. Available at https://github.com/MicroGSD/RoadArchitect/

[6] Research Group: Mobile Interactive Systems: AutoWSD VR Driving Simulator. Available at https://mint.fh-hagenberg.at/

All the questionnaires use 7-point Likert scales ranging from 1 (strongly disagree) to 7 (strongly agree)

4 User Study

4.1 Participants and Procedure

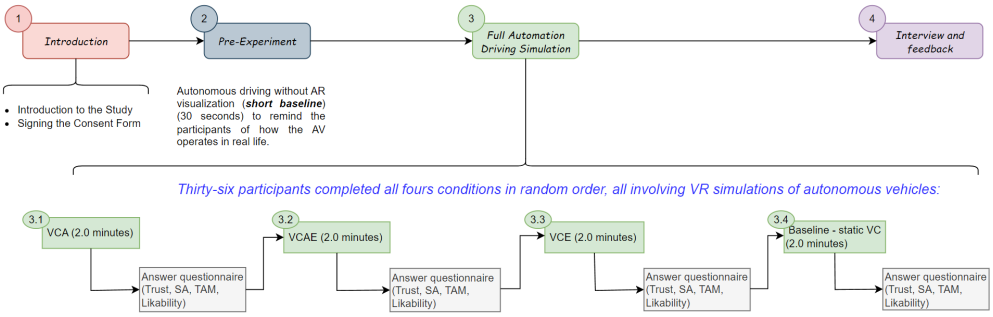


Fig. 3: Procedure of the user study

The number of participants for an ANOVA repeated measure analysis was determined via an a priori power analysis using G*Power, with an alpha level of 0.05. Twenty-eight participants should have a medium effect size of 0.25. We recruited 36 participants from members of the laboratory and other university students, with 23 males and 13 females aged from 20 to 53 (mean=30.36, SD=10.37). On a 7-point Likert scale ranging from 1 (Strongly Disagree) to 7 (Strongly Agree), participants showed medium trust in AVs (M=4.6, SD=1.4). Of the participants, 83.3% had a driver's license.

Below is an outline of the experimental procedure (see Figure 3):

1. Introduction: Participants begin the study with a brief introduction in which the researcher explains the overall purpose and structure of the experiment. They then read and signed the consent form to indicate informed and voluntary participation. Participants subsequently completed a demographic questionnaire that collected information on age, gender, driver's license, and self-reported trust in AVs. Trust was assessed using a 7-point Likert scale in response to the statement "I trust autonomous vehicles".
2. Pre-Experiment (Short Baseline): Next, participants were instructed to sit in a chair (the passenger seat) and to adjust the Oculus Quest Pro headset. They complete a short 30-second autonomous driving without any AR visualization. The goal is to remind participants of how an AV typically operates.
3. Full Automation Driving Simulation: Participants then proceed to the main part of the study, which consists of four different VR conditions presented in randomized order. Each condition lasts two minutes and features a specific type of VC behavior: VCA, VCAE, VCE, and a static VC (baseline). After each condition, participants complete a questionnaire measuring trust, SA, TAM, and Likability

(detailed in Section 3.5), providing insight into how each visualization influences their perceptions. The questionnaire is shown in a VR headset.

4. Interview and Feedback: Participants take part in a final interview session. This qualitative phase allows them to share their impressions, preferences, and any difficulties they experienced during the simulations.

The total duration of the experiment was approximately 40–45 minutes.

We focused on AVs with the highest level of automation (level 5 [8]). In these AVs, vehicles are entirely automated and may lack traditional controls like pedals and steering wheels. Additionally, we aimed for participants not to control the car but to only observe the environment. To this end, participants sat in the passenger seat.

4.2 Informed consent

The study had the approval of the university ethics committee. All participants provided informed consent before participating in the study. Participation was voluntary, and participants could withdraw at any time.

4.3 Results

In exploring the impact of different conditions on trust, TAM, SA, and Likability, outliers were identified, the Friedman Test was employed, and then the Wilcoxon signed-rank test was used for post-hoc analysis. P-values were adjusted using the Bonferroni correction. Ggboxplot [58] was used to generate the figures. The figures include box plots showing data distribution, and also include statistical details: the test used, chi-square, p-value, effect size, and sample size.

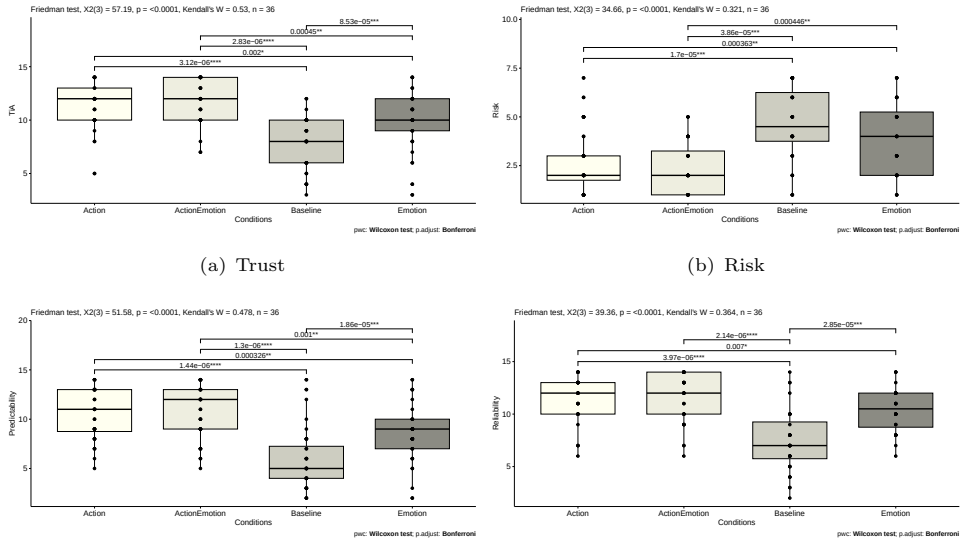
To examine the interaction of Action and Emotions on trust, SA, TAM, or likability, we first performed an Aligned Rank Transform (ART) for non-parametric factorial data analysis [59]. A two-way ANOVA analysis was then performed.

4.3.1 Trust

The trust questionnaire includes four sub-factors: trust in automation, perceived risk, predictability, and reliability.

Friedman's test found statistically significant differences in the different conditions for trust ($X^2(3) = 57.19, p < 0.0001, W = 0.53, 95\%CI[0.39, 0.69]$), Perceived Risk ($X^2(3) = 34.66, p < 0.0001, W = 0.32, 95\%CI[0.21, 0.5]$) Predictability ($X^2(3) = 51.58, p < 0.0001, W = 0.48, 95\%CI[0.32, 0.64]$), and Reliability ($X^2(3) = 39.36, p < 0.0001, W = 0.36, 95\%CI[0.24, 0.54]$).

Follow-up Wilcoxon signed-rank tests indicated significant pairwise differences between conditions (see Figure 4). All AR-based VC conditions (VCA, VCE, and VCAE) produced significantly higher scores in trust, predictability, and reliability relative to the baseline (see Figures 4(a), 4(c), and 4(d)). For perceived risk, both VCA and VCAE showed significantly lower risk ratings than the baseline (see Figure 4(b)). Furthermore, VCA and VCAE yielded significantly better outcomes than VCE across all four trust-related sub-factors (see Figure 4).



(a) Trust (b) Risk

(c) Predictability (d) Reliability

Fig. 4: Results of Friedman's tests for trust factors

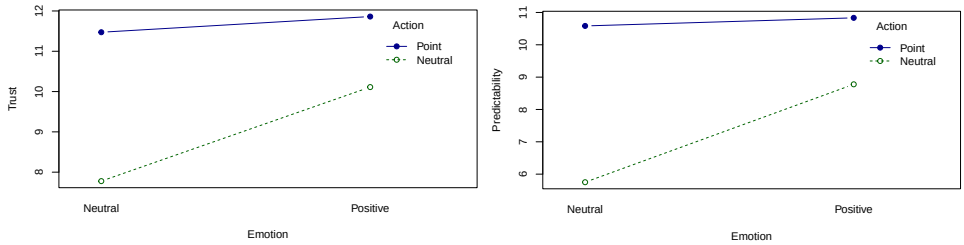
After using ART for non-parametric factorial data, we conducted a two-way ANOVA. The results showed a significant interaction effect (IE) of Action x Emotion on trust, reliability, and predictability (see Figure 5). It means that the effect of Action on trust, Reliability, and Predictability depends on the level of Emotion and vice versa. Furthermore, there was a significant main effect of Action on trust, reliability, and predictability (see Table 1), indicating that Action directly affects these variables. Similarly, results identified a significant main effect of Emotion on trust, reliability, and predictability (see Table 1), signifying that Emotion also plays a role in influencing these factors. The partial Eta-squared (η_p^2) of the Action factor is larger than the Emotion factor (see Table 1).

	Trust				Predictability				Reliability			
	F(1, 105)	p-value	η_p^2	95%CI	F(1, 105)	p-value	η_p^2	95%CI	F(1, 105)	p-value	η_p^2	95%CI
Action	81.03	< 0.0001	0.43	[0.32,1]	83.01	< 0.0001	0.44	[0.33,1]	59.64	< 0.0001	0.36	[0.25,1]
Emotion	20.04	< 0.0001	0.16	[0.07,1]	19.17	< 0.0001	0.15	[0.06,1]	23.72	< 0.0001	0.18	[0.09:1]
Action:Emotion	8.40	< 0.005	0.07	[0.01,1]	13.53	< 0.001	0.11	[0.04,1]	19.85	< 0.0001	0.16	[0.07:1]

Table 1: Results of two-way ANOVA on trust factor

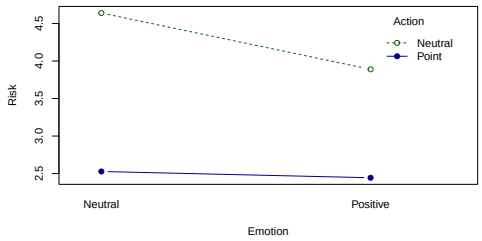
4.3.2 Situational Awareness

The SART questionnaire includes two sub-factors: Supply (perceived workload) and Understandable. Friedman's test is used to assess the impact of different conditions on Supply and Understandable. The results found significant differences in the different conditions for SART sub-scale Understandable ($X^2(3) = 61.99, p < 0.0001, W =$

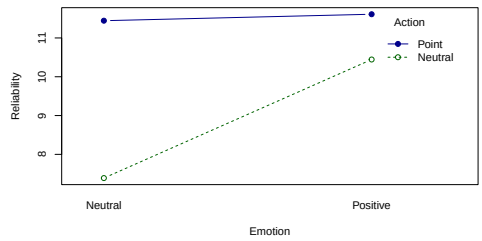


(a) Trust

(b) Predictability

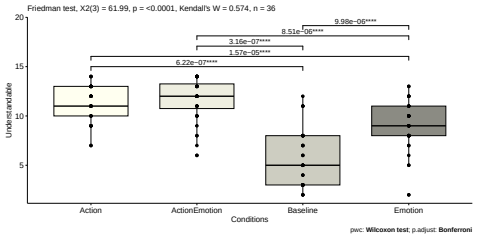


(c) Risk

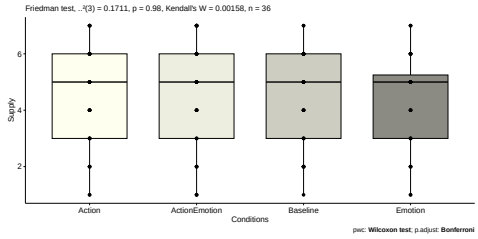


(d) Reliability

Fig. 5: Results of interaction effect on trust factors



(a) Understanding of the provided situation



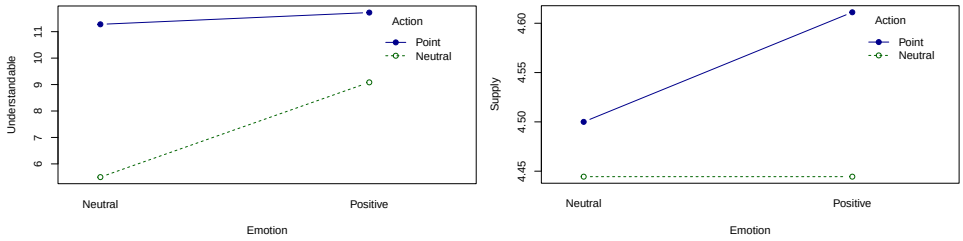
(b) Supply (perceived workload)

Fig. 6: Results of Friedman's test for SART

$0.57, 95\%CI[0.41, 0.73]$). However, no statistically significant difference was observed in the SART sub-scale Supply (see Figure 6(b)).

Pairwise Wilcoxon signed rank test revealed statistically significant differences in Understandable (see Figure 6). Understandable was rated significantly higher when using the AR technique to add a VCA, VCE, or VCAE than the baseline. Moreover, statistically significant differences were observed between the 'Action' and 'Action Emotion' conditions when compared to the 'Emotion' condition (see Figure 6(a)).

After the data were transformed with ART, we conducted a two-way ANOVA. The results found a significant IE of Action x Emotion on Understandable (see Figure 7). This suggests that the impact of Action on Understandable varies depending on the level of Emotion, and vice versa. Furthermore, a significant main effect of Action on Understandable was observed, indicating that Action directly influences Understandable. Similarly, a significant main effect of Emotion on Understandable was identified



(a) SA-Understandable (b) SA-Supply

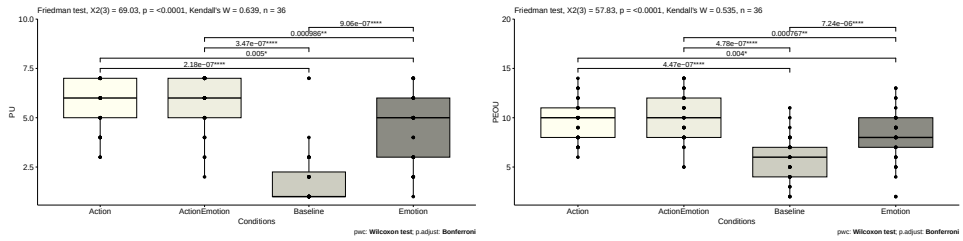
Fig. 7: Results of interaction effect on SART

(see Table 2), suggesting that Emotion also plays a role in affecting Understandable. The η_p^2 of the Action factor is larger than the Emotion factor (see Table 2). There was no main effect or IE of Action or Emotion on Supply.

	Understandable				PEOU				PU			
	F(1, 105)	p-value	η_p^2	95%CI	F(1, 105)	p-value	η_p^2	95%CI	F(1, 105)	p-value	η_p^2	95%CI
Action	142.1	< 0.0001	0.57	[0.48,1]	82.48	< 0.0001	0.44	[0.33,1]	103.18	< 0.0001	0.59	[0.5,1]
Emotion	36.77	< 0.0001	0.26	[0.15,1]	19.61	< 0.0001	0.16	[0.06,1]	19.74	< 0.0001	0.34	[0.22,1]
Action:Emotion	23.46	< 0.0001	0.18	[0.08,1]	10.62	0.001	0.1	[0.02,1]	2.01	0.16	0.29	[0.18,1]

Table 2: Results of two-way ANOVA on SA-Understand, TAM-PEOU, and TAM-PU

4.3.3 Technology Acceptance Model

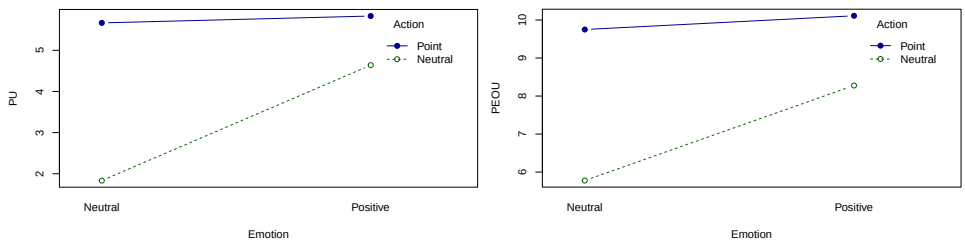


(a) Perceived Usefulness (b) Perceived Ease of Use

Fig. 8: Results of Friedman's test for TAM

The TAM questionnaire includes two sub-factors: PEOU and PU. Friedman's test is used to assess the impact of different conditions on TAM-PU and TAM-PEOU.

Friedman's analysis found statistically significant differences in the different conditions for PEOU ($X^2(3) = 57.83, p < 0.0001, W = 0.54, 95\%CI[0.38, 0.68]$). Pairwise Wilcoxon signed rank test revealed statistically significant differences in PEOU (see Figure 8(b)). The conditions for the PEOU can be ranked as follows: best VCA and VCAE, next is VAE, and last is the baseline. The two-way ANOVA results found a significant IE of Action x Emotion on TAM-PEOU (see Figure 9). 'Action' also



(a) TAM - Perceived Usefulness (b) TAM - Perceived Ease of Use

Fig. 9: Results of interaction effect on TAM

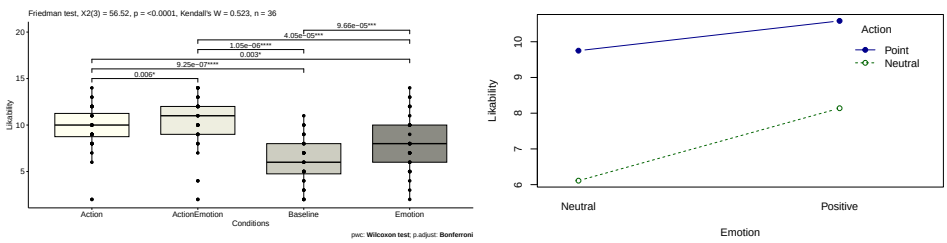
has a significant impact on TAM-PEOU. Similarly, a two-way ANOVA indicated a significant main effect of Emotion on TAM-PEOU (see Table 2).

Friedman's test found statistically significant differences in the different conditions for TAM-PU ($X^2(3) = 69.03, p < 0.0001, W = 0.64, 95\%CI[0.52, 0.77]$). Pairwise Wilcoxon signed rank test revealed statistically significant differences in PU (see Figure 8(a)). The Usefulness for each of the conditions can be ranked as follows: the best is VCA and VCAE, the next best is VCE, and the least useful is the baseline. After using ART for non-parametric factorial data, we conducted a two-way ANOVA. The results found a significant IE of Action x Emotion on TAM-PU (see Figure 9). The effect of Action on TAM-PU depends on the level of Emotion and vice versa. There is also a significant main effect of Action on TAM-PU (see Table 2). Similarly, results indicated a significant main effect of Emotion on TAM-PU (see Table 2).

Based on the η_p^2 and p-value (η_p^2 of Action larger than Emotion, p-value of Action smaller than Emotion), Action has more impact on TAM-PEOU and TAM-PU (see Table 2).

PU and PEOU are rated significantly higher using AR to add a VCA, VCAE, or VCE compared to the baseline scenario (see Figure 8). Additionally, statistically significant differences are observed between the conditions of a VCA and VCAE compared to the VCE, in both PEOU and PU.

4.3.4 Likability



(a) Results of Friedman's tests on Likability (b) Results of two-way ANOVA tests on Likability

Fig. 10: Results of ANOVA tests for Likability

The results of Friedman's analysis showed statistically significant differences in the different conditions for Likability ($X^2(3) = 56.52, p < 0.0001, W = 0.52, 95\%CI[0.35, 0.69]$). Pairwise Wilcoxon signed rank test revealed statistically significant differences in Likability (see Figure 10). The VCAE is rated significantly higher than other conditions in terms of Likability. The Likability of the baseline condition is rated significantly lower than in other scenarios (see Figure 10(a)).

Likability	F(1,105)	p-value	η_p^2	95%CI
Action	103.18	< 0.0001	0.5	[0.39,1]
Emotion	19.74	< 0.0001	0.16	[0.07,1]
Action:Emotion	2.01	0.16	0.02	[0.01,1]

Table 3: Results of two-way ANOVA tests for Likability

The data was transformed with the ART method; we then conducted a two-way ANOVA. The result found no significant IE of Action x Emotion on Likability (see Figure 10(b)). However, there is a significant main effect of Action and Emotion on Likability (see Table 3). Both main effects are significant; without a significant IE means that both factors (Action and Emotion) independently influence Likability.

4.3.5 Individual Feedback

All participants found the virtual character's actions very helpful. When it pointed towards potential hazards or changes in the environment, it made them more aware of their surroundings. Most of the participants (29 out of 36) agreed that the virtual character's emotional expressions were reassuring. When the virtual character displayed a smiling expression, it made them feel less stressed, even when navigating through complex or busy environments. The VC gave them a sense of the vehicle's confidence in managing the situation. The participants also agreed that combining both emotion and action was very effective. The VC's actions provided clear guidance, while its emotions gave participants a sense that the vehicles could manage the situation.

5 Discussion

Taken together, our results demonstrate that using AR to add a VC with action (VCA), emotion (VCE), or both (VCAE) enhances trust, predictability, and reliability compared to a baseline without action or emotion. Specifically, participants in conditions with VCs (VCE, VCA, VCAE) reported greater confidence in the AV's abilities and found the vehicle's behavior more predictable.

Impact on Trust, these findings support **H1**, as all VC-enhanced conditions surpassed the baseline in fostering user trust. In these scenarios, the VC serves as a representative entity of the AV and its AI component, enabling passengers to interpret the vehicle's environmental awareness. Notably, even VCs that solely conveyed emotion (VCE) enhanced trust, predictability, and reliability, suggesting that positive emotional cues alone can communicate the AV's responsiveness. For example, participants interpreted the VC's smile as indicating that the AV was aware of a potential

hazard, prompting them to expect a safety response, such as deceleration. Conversely, in the baseline condition with a static VC, participants expressed confusion and concern over the AV's actions, especially when it slowed down without any visible cues, leading them to feel less in control and informed.

Situational Awareness (SA), the results reveal that VCs exhibiting only action (VCA), those solely displaying emotion (VCE), and those combining both action and emotion (VCAE) contribute to an enhanced understanding of the environment and potential hazards. Thus, **H2** is supported, as all three VC configurations improved participants' SA relative to the baseline. When focused on the environment, participants were quickly drawn to the VC's action, particularly when it pointed to specific areas of concern. This directed attention allowed participants to immediately identify potential dangers, whereas emotion-only VCs required them to spend time scanning the environment to locate the hazard. Thus, actions were critical in directly enhancing environmental awareness. Notably, these benefits were achieved without increasing perceived cognitive workload, as evidenced by no statistically significant change in Supply sub-scale scores.

The Technology Acceptance Model (TAM), VCE achieved significantly higher scores than the baseline in both perceived usefulness and perceived ease of use, supporting **H3**. Participants noted that the positive emotional expressions of the VC helped reduce their stress, particularly in situations involving environmental uncertainty. Emotion-based VCs, under their social presence, provided a comforting presence. However, the VCA condition achieved even higher TAM scores, suggesting that action plays a more impactful role than emotion alone in fostering ease of use and perceived usefulness. Analysis of interaction effects further supports this, indicating that action exerts a stronger influence than emotion on trust, SA, and TAM.

Our results advance the existing literature by extending and differentiating from the findings of Häuslschmid et al. [10] and Chakravarthi Kumaran et al. [11] in several ways. Häuslschmid et al. demonstrated that a chauffeur avatar presented on a HUD could modestly increase comfort but showed no significant improvement over baseline in terms of trust, in part because the avatar provided only minimal behavioral cues (e.g., rear-facing glances) and lacked visibility of facial expressions. In contrast, our study employed a VC capable of delivering animated, front-facing actions and emotions during multiple in-ride uncertain driving events. This richer expressive capacity enabled an examination of how specific behavioral modalities—actions, emotions, and their combination—shape trust and SA.

Similarly, Chakravarthi Kumaran et al. [11] showed that a non-humanoid physical robot could improve trust during pre-trip interactions, but its abstract form and limited movement prevented the expression of distinct emotional or context-specific action cues. Whereas the robot in their study acted only before the journey and offered static social presence, our VC actively responded to unfolding environmental cues, enabling a deeper assessment of how passengers interpret system awareness during the trip.

However, our results also diverge from prior work emphasizing the power of emotional expressiveness in virtual agents [33, 35]. In contrast to studies showing that positive emotions (e.g., smiling) enhance perceived trustworthiness or credibility, emotional expressions alone in our VC produced smaller effects than action cues. Emotional

cues in isolation may lack contextual grounding. Participants may have found it difficult to interpret an emotional reaction without a corresponding action that clarified the cause of the emotion.

Interestingly, the combination of action and emotion (VCAE) achieved the highest likability ratings among all conditions. Participants responded positively to the VC's human-like qualities when both action and emotion were present, which led to a stronger overall connection and enhanced user experience. This suggests that an emotionally expressive and action-capable VC not only improves functional aspects such as SA and trust but also fosters a more enjoyable, engaging interaction.

Overall, the findings suggest that the primary objective of adding a VC is to effectively capture user attention in uncertain situations, where both actions and emotional expressions help clarify the AV's awareness and intentions. VCs that highlight uncertainty in the environment give users time to comprehend the situation and predict the AV's response. This study suggests that a well-designed VC should use clear, easily understood actions and emotions to avoid ambiguity and enhance user experience. To maximize their impact, VCs should:

- Use clear, intuitive gestures to convey environmental awareness and AV intentions.
- Incorporate emotional expressions, such as smiles, to reassure passengers during uncertain situations or to show that the AV has identified the uncertain situation.
- Balance anthropomorphic design without triggering the "uncanny valley" effect.

6 Limitations and Future Work

This study has several limitations.

First, although prior work indicates that VR environments can effectively simulate AV driving, the sense of safety they provide may not fully replicate real-world conditions. As trust is closely tied to risk perception and the alignment between system behavior and actual hazard mitigation [60, 61]. This reduced risk may have inflated trust ratings or diminished differences between experimental conditions. Future studies should increase the realism of simulated AV scenarios, for example, by incorporating near-miss events, to generate a greater sense of potential danger. Assessing how these effects transfer to real-world AV contexts will be an important next step.

Second, the VR system used in this study accurately detected uncertain situations and conveyed them to the VC without delay or error. In real AVs, however, misclassifications, late detections, or missed hazards are common and directly influence trust formation, particularly when the system fails to behave in accordance with user expectations. Prior work has shown that self-reported trust decreases sharply following automation malfunctions [62]. Future research should incorporate controlled system errors, including misrecognitions, delays, or missed detections, to examine how VC actions and emotional cues might dynamically recalibrate trust under imperfect real-world conditions.

Third, the duration of each VR trip was relatively short. Future studies should investigate the impact of the VC on trust during the performance phase, which focuses on long-term usage [24].

Fourth, the participant sample was not fully representative of the broader population. Most participants (83.3%) held a driving license, and the age range was concentrated between 20 and 53 years. The absence of older adults, non-drivers, or individuals with limited driving experience limits the generalizability of the findings. Broader demographic sampling will be essential for understanding how different user groups perceive and respond to VCs in AVs.

Fifth, the experiment incorporated a limited set of risk scenarios, focused on pedestrians crossing and unmarked intersections. More complex or adverse driving environments—such as nighttime driving, severe weather, or multi-hazard scenarios—were not included. Expanding the scenario set will help determine how well VCs support passengers under more challenging and dynamic AV conditions

A further limitation concerns the conceptual boundaries among the constructs measured in this study. Although validated instruments such as TAM, SART, and trust questionnaires were used, their combined use may lead to partial construct overlap, making it difficult to disentangle the unique contribution of the VC to each outcome. Future research should employ a more explicit theoretical framework that clarifies expected relationships among these constructs:

A limitation of our design is the use of a static VC as the baseline. While this choice ensured visual consistency, it prevented us from assessing how participants might respond to autonomous driving without any VC present. Future work should therefore include a no-VC baseline to enable direct comparison with VC conditions.

An additional consideration concerns the feasibility of deploying such a VC in real vehicles. In our VR environment, the VC's position and pointing behaviors were synchronized with simulated hazards; however, rendering and anchoring a VC in a real vehicle introduce substantial technical challenges. Accurate spatial alignment would require robust AR anchoring, real-time object tracking, and low-latency integration of perception and rendering pipelines to ensure that the VC's gestures remain spatially correct relative to moving hazards. Future work should investigate how emerging AR frameworks and perception systems can support stable rendering and spatial accuracy for in-ride VCs.

Finally, the study employed a single VC design with predefined actions and emotions. The exclusion of negative emotions may have reduced the expressive range of the VC, and participants noted that emotions such as surprise or fear might feel more natural, like human behavior in certain uncertain situations. Future research should examine a wider spectrum of emotional intensities and varied degrees of anthropomorphism to better understand how VC design influences trust and acceptance. Developing a standardized list of VC actions and emotions tailored to distinct driving scenarios will also be important.

7 Conclusion

In this paper, we propose a design for a VC aimed at enhancing trust in AVs. We evaluate trust in AVs based on two characteristics of the VC: its emotional expression and its actions.

The primary goal of the VC is to serve as an intuitive and easily understandable interface that informs passengers about the AV's awareness of its surroundings and its ability to take appropriate actions in uncertain situations. We conducted a user study involving four AV trips with four different VC conditions: a VC with emotion only, a VC with action only, a VC with both action and emotion, and a baseline condition with a static VC. Across these conditions, we compare the VC's action and emotion impact on trust, Situation Awareness, Technology Acceptance, and Likability.

The findings indicate that the inclusion of a VC enhances trust, Situation Awareness, Technology Acceptance, and Likability in AVs. The VC's emotional expressions reduce stress among participants during uncertain situations; the emotion serves as a signal to tell passengers that the AV has the ability to manage the situation. This signal aids users in better comprehending the situation, predicting, and understanding the vehicle's intentions. However, the VC's actions are more effective in conveying the AV's intentions and actions, capturing passengers' attention more efficiently and aiding in rapid understanding of the situation. In contrast, with only emotional cues, passengers must identify hazardous conditions within the environment themselves. In conclusion, VCs exhibiting both emotional expressions and actions were rated highest in likability. The emotional cues from the virtual characters reduced stress, while their actions effectively conveyed the AV's performance.

Supplementary information. Below is the link to the electronic supplementary material: https://github.com/ThanhHoa97/VirtualCharacter/ in the Material folder

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Declarations

- Funding: IMT Atlantique and University of South Australia
- Conflict of interest/Competing interests (check journal-specific guidelines for which heading to use): The authors declare no conflict of interest.
- Ethics approval and consent to participate: The study had the approval of the university ethics committee.
- Consent for publication: The authors affirm that participants provided informed consent for publication
- Data availability: The data presented in this study are openly available on https://github.com/ThanhHoa97/VirtualCharacter/ in the Data folder.
- Materials availability: The material needed for this study is available at https://github.com/ThanhHoa97/VirtualCharacter.git
- Code availability: The code used for this study is available on GitHub at https://github.com/ThanhHoa97/VirtualCharacter.git
- Author contribution: Conceptualization, TTH.T., E.P., J.W., G.M., and B.H.T. Data curation, TTH.T. Methodology, TTH.T., E.P., J.W., G.M., and B.H.T. Supervision, E.P., J.W., G.M., and B.H.T. Writing–review and editing, TTH.T., E.P., J.W., G.M,. and B.H.T.

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